Lecture 20: Diagonalizing a matrix

$A = S\Lambda S^{-1}$

Powers of A

Equation $u_{k+1} = A \cdot u_k$

Last time: $Ax = \lambda x$

$A - \lambda I$ singular

Suppose A has n indep. eigenvectors. Put them in the columns of S.

$$AS = A \cdot \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{bmatrix} = \begin{bmatrix} | & | & & | \\ \lambda_1 x_1 & \lambda_2 x_2 & \cdots & \lambda_n x_n \\ | & | & & | \end{bmatrix}$$

$$= \begin{bmatrix} | & | & & | \\ x_1 & x_2 & \cdots & x_n \\ | & | & & | \end{bmatrix} \underbrace{\begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ & & & \lambda_n \end{bmatrix}}_{\text{diagonal eigenvalue matrix } \Lambda} = S\Lambda$$

$AS = S\Lambda \implies S^{-1}AS = \Lambda$

$A = S\Lambda S^{-1}$

If $Ax = \lambda x$, then

$A^2 x = A(\lambda x) = \lambda A x = \lambda^2 x.$

$A^2 = S\Lambda S^{-1} \cdot S\Lambda S^{-1}$

$= S\Lambda^2 S^{-1}$

$A^k = S\Lambda^k S^{-1}$

<u>Theorem</u>: $A^k \to 0$ as $k \to \infty$ if $|\lambda_i| < 1$ for all i.

A is sure to have n indep eigenvalues (and be diagonalizable) if all λ's are distinct.

Repeated λ's: may or may not have n indep eigenvectors.

<u>Ex</u>: $A = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix}$

$$\det(A - \lambda I) = \begin{vmatrix} 2-\lambda & 1 \\ 0 & 2-\lambda \end{vmatrix} = (2-\lambda)^2 = 0$$

$\lambda = 2, 2.$

$$\left[\begin{array}{cc|c} 0 & 1 & 0 \\ 0 & 0 & 0 \end{array}\right] \rightsquigarrow x = c\begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Equation $u_{k+1} = A \cdot u_k$

Start with given vector u_0

$u_1 = Au_0, \quad u_2 = A^2 u_0, \cdots, \boxed{u_k = A^k u_0}$

To really solve: write

$$u_0 = c_1 x_1 + c_2 x_2 + \cdots + c_n x_n = Sc$$

$$A^{100} u_0 = c_1 \lambda_1^{100} x_1 + c_2 \lambda_2^{100} x_2 + \cdots + c_n \lambda_n^{100} x_n.$$

$$= \Lambda^{100} Sc$$

$$= u_{100}$$

Fibonacci: 1, 1, 2, 3, 5, 8, 13, 21, 34, ...

$\boxed{F_{100} = ?}$

$$\begin{cases} F_{k+2} = F_{k+1} + F_k \\ F_{k+1} = F_{k+1} \end{cases}$$

<u>Trick</u>: $u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$

$$\begin{bmatrix} F_{k+2} \\ F_{k+1} \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix}$$

$$u_{k+1} = A \cdot u_k$$

$$\det(A - \lambda I) = \begin{vmatrix} 1-\lambda & 1 \\ 1 & -\lambda \end{vmatrix} = (1-\lambda)(-\lambda) - 1$$

$$= \lambda^2 - \lambda - 1 = 0$$

$$\Rightarrow \lambda = \frac{+1 \pm \sqrt{1+4}}{2}$$

$\lambda_1 = \frac{1}{2}(1+\sqrt{5}) \cong 1.618\ldots \qquad x_1 = \begin{bmatrix} \lambda_1 \\ 1 \end{bmatrix}$

$\lambda_2 = \frac{1}{2}(1-\sqrt{5}) \cong -0.618\ldots \qquad x_2 = \begin{bmatrix} \lambda_2 \\ 1 \end{bmatrix}$

$$u_0 = \begin{bmatrix} 1 \\ 1 \end{bmatrix} = c_1 x_1 + c_2 x_2$$

solve for c

$$u_k = \begin{bmatrix} F_{k+1} \\ F_k \end{bmatrix} = c_1 \lambda_1^k \begin{bmatrix} 1 \\ \lambda_1 \end{bmatrix} + c_2 \lambda_2^k \begin{bmatrix} 1 \\ \lambda_2 \end{bmatrix}$$

$$F_{k+1} = c_1 \lambda_1^k + c_2 \lambda_2^k \approx c_1 \lambda_1^k$$